[62]
0s

```python
if "Item_Name" in df_fe.columns:
    top_k =3
    top_items = df_fe["Item_Name"].value_counts().head(top_k).index
    df_fe["Item_Name_reduced"] = np.where(df_fe["Item_Name"].isin(top_items), df_fe["Item_Name"], "Other")
    print("Unique Item_Name:", df_fe["Item_Name"].nunique())
    print("Unique Item_Name_reduced:", df_fe["Item_Name_reduced"].nunique())
    df_fe[["Item_Name","Item_Name_reduced"]].head(10)
else:
    print("Item_Name column not found.")
```

```
Unique Item_Name: 9
Unique Item_Name_reduced: 4
```

	feature	importance
38	price_per_item	0.084617
33	Delivery_Distance_km	0.084054
30	Customer_Lon	0.082394
31	Driver_Lat	0.082239
32	Driver_Lon	0.082081
27	Restaurant_Lat	0.081720
29	Customer_Lat	0.081719
28	Restaurant_Lon	0.081447
34	order_hour	0.057204
35	order_dayofweek	0.041031
26	Quantity	0.033049
20	Item_Name_reduced_Other	0.011978
7	Payment_Method_Cash	0.011534
11	Driver_Vehicle_Car	0.011501
9	Payment_Method_Wallet	0.011270

[41]
22s

```python
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred), 4))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

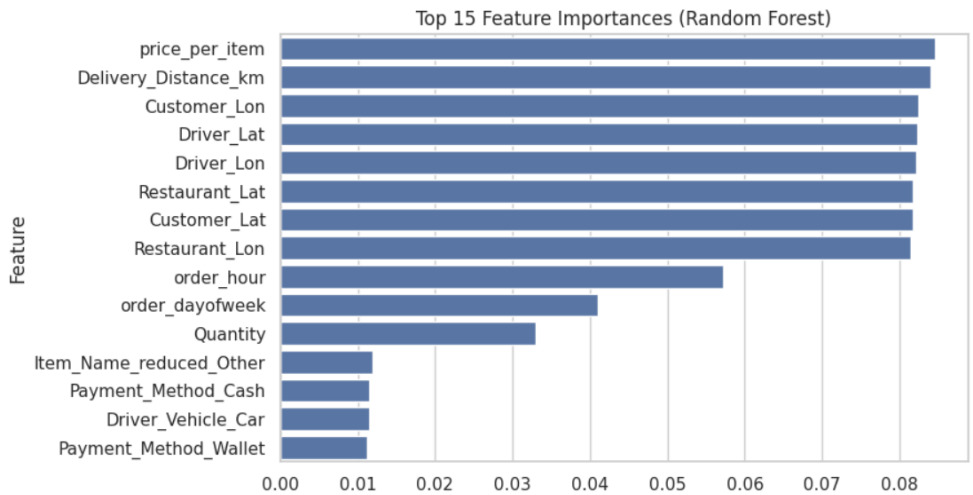
```
Accuracy: 0.4008

Classification Report:
              precision    recall  f1-score   support

   Cancelled       0.39      0.46      0.42      1963
   Delivered       0.41      0.54      0.46      2039
  In Transit       0.10      0.00      0.00       998

    accuracy                           0.40      5000
   macro avg       0.30      0.33      0.30      5000
weighted avg       0.34      0.40      0.36      5000
```

```
[44]
sns.barplot(data=fi.head(top_n), x="importance", y="feature")
plt.title(f"Top {top_n} Feature Importances (Random Forest)")
plt.xlabel("Importance")
plt.ylabel("Feature")
plt.show()
```




```
[73]
if "Item_Name" in df_fe.columns:
    top_k =5
    top_items = df_fe["Item_Name"].value_counts().head(top_k).index
    df_fe["Item_Name_reduced"] = np.where(df_fe["Item_Name"].isin(top_items), df_fe["Item_Name"], "Other")
    print("Unique Item_Name:", df_fe["Item_Name"].nunique())
    print("Unique Item_Name_reduced:", df_fe["Item_Name_reduced"].nunique())
    df_fe[["Item_Name","Item_Name_reduced"]].head(10)
else:
    print("Item_Name column not found.")
```

```
Unique Item_Name: 9
Unique Item_Name_reduced: 6
```

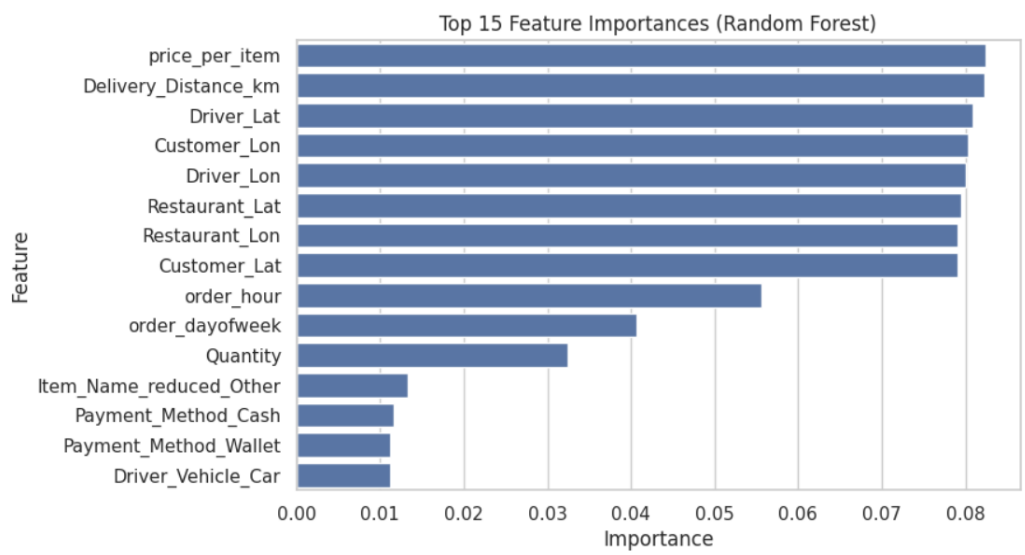
```
fi.head(15)
```

	feature	importance
40	price_per_item	0.082463
35	Delivery_Distance_km	0.082318
33	Driver_Lat	0.080867
32	Customer_Lon	0.080348
34	Driver_Lon	0.080023
29	Restaurant_Lat	0.079558
30	Restaurant_Lon	0.079118
31	Customer_Lat	0.079113
36	order_hour	0.055587
37	order_dayofweek	0.040729
28	Quantity	0.032372
20	Item_Name_reduced_Other	0.013335
7	Payment_Method_Cash	0.011627
9	Payment_Method_Wallet	0.011276
11	Driver_Vehicle_Car	0.011198

[63]
✓ 20s

```python
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred), 4))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.3882

Classification Report:
              precision    recall  f1-score   support

   Cancelled       0.38      0.45      0.42      1963
   Delivered       0.39      0.52      0.45      2039
  In Transit       0.06      0.00      0.00       998

    accuracy                           0.39      5000
   macro avg       0.28      0.32      0.29      5000
weighted avg       0.32      0.39      0.35      5000
```